

Plant Science Brief: Drought Tolerance in Wheat

Drought is the single largest abiotic constraint on wheat (*Triticum aestivum*) productivity worldwide, with global yield losses estimated at 10-50% in affected seasons depending on developmental stage and severity. This brief summarises canonical molecular and physiological mechanisms cereal crops deploy to tolerate water deficit, with a focus on wheat.

Hormonal Signalling

Abscisic acid (ABA) is the master regulator of drought response. Stress triggers ABA biosynthesis primarily through NCED gene family upregulation. ABA binds PYR/PYL/RCAR receptors, inhibits PP2C phosphatases, and allows SnRK2 kinases to phosphorylate downstream effectors including SLAC1, AREB/ABF transcription factors, and ion channels.

Stomatal Regulation

ABA-driven stomatal closure reduces transpirational water loss but also constrains CO₂ uptake. Cultivars with rapid stomatal closure show improved early-stage survival; cultivars that maintain partial stomatal conductance perform better under mild, sustained deficit.

Osmotic Adjustment

Accumulation of compatible solutes -- proline, glycine betaine, trehalose, and soluble sugars -- maintains cell turgor under low water potential. P5CS and BADH gene family overexpression has been demonstrated to improve drought tolerance in transgenic wheat and barley.

Key Loci

- TaSnRK2.10 -- kinase central to ABA signalling; linked QTLs on chromosome 6A associate with field drought tolerance.
- TaDREB1A -- DREB/CBF-family transcription factor binding DRE/CRT cis-elements upstream of stress-response genes.
- TaNAC69 -- NAC family; ABA-responsive, regulates root architecture under deficit.

This brief is a synthesised demo summary and not a peer-reviewed source.